

William Kwako

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EDUCATION

M.S., Computer Science – Interactive Intelligence, Georgia Tech, Atlanta, GA

August 2023 - August 2026

B.A., Physics, Willamette University, Salem, OR

May 2019

PROFESSIONAL EXPERIENCE

Technical Solutions Engineer – Epic Systems Corporation

December 2019 – December 2022

- Supported deployment, maintenance, and troubleshooting of revenue cycle software, including claims processing and electronic remittance, for 5 health system clients across the US and Canada
- Headed the Debugging Task Force; taught new hires to use internal debugging tools and revised the onboarding curriculum
- Wrote M/MUMPS database queries against production healthcare IT databases to diagnose client issues and support data analysis for issue resolution
- Coordinated implementation of large-scale claims features for clients, including claim attachments, PLBs, rapid reconciliation, and state reporting

Teacher's Assistant – Willamette University, Physics Department

January 2019 – May 2019

- Guided students through computational and experimental Introductory Physics 2 laboratory exercises
- Answered questions about class material and evaluated students' performance

Intern – Willamette University, Physics Department

May 2018 – July 2018

- Implemented 3D wavelet convolutions via FFT in Python to search for baryon acoustic oscillation (BAO) signals in the 2MASS Redshift Survey – the first application of this method to the 2MRS
- Applied Gaussian smoothing, zero-padding, and a custom local maxima detection algorithm to identify candidate BAO locations from a cosmological density field
- Built a computer for parallel GPU processing to support analysis of a large-scale cosmological dataset

PROJECTS AND RESEARCH

- **Benchmarking Reinforcement and Imitation Learning on a Deterministic Platformer: Jump King**
Reinforcement Learning Research, 2026
 - Built a full pipeline interfacing directly with a live game via C# memory extraction, named pipe communication, and a custom Gymnasium environment
 - Designed and implemented behavioral cloning (BC) from human playthroughs, per-level reward shaping, and infrastructure to rigorously compare RL, BC, and BC+RL approaches
 - Agents reliably complete 100% of the base game (43 chained levels) in ~10 minutes
 - **Tools:** Python, C#, Stable-Baselines3, Gymnasium, dnSpy, .NET
- **Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems**
Educational Technology Research, 2025-2026
 - Built a web application generating personalized coding exercises to improve code comprehension, a skill fundamental to programming but rarely practiced deliberately

- Designed and implemented a multi-model agentic pipeline with conditional routing, prompt chaining, input validation, guardrails, and feedback loops across OpenAI and Anthropic APIs
- **Tools:** Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs
- **Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation**
Cosmology Research – Senior Thesis, Willamette University, 2018-2019
 - Applied wavelet transformations to identify baryon acoustic oscillations (BAOs) in galaxy distribution data from the 2MASS Redshift Survey (2MRS)
 - Identified three BAO signals with only 0.0008% probability of occurring by chance (3.41σ), corroborating and extending results of prior studies
 - **Tools:** Python
- **Schema Activation via Source Attribution: Does Labeling a Passage as AI-Authored Affect Immediate Recall?**
Cognitive Science Research, 2026
 - Investigated whether labeling a passage as AI-authored activates a source-credibility schema that affects immediate memory recall
 - Recruited 22 participants across three experimental groups exposed to identical passages attributed to different sources
 - Participants in the AI-labeled group scored nearly 17 percentage points higher on recall than other groups, suggesting low-credibility, high-skepticism schema activation

POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*
Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
Oral Presentation – Willamette-Linfield intercollegiate symposium (May 2019)
Poster Presentations – 223rd Meeting of the American Astronomical Society (January 2019),
Murdock College Science Research Conference (November 2018)

TECHNICAL SKILLS

- **Languages:** Python, HTML, CSS, JavaScript, SQL, M/MUMPS
- **AI & ML:** NumPy, Stable-Baselines3, Gymnasium, OpenAI API, Anthropic API, Prompt Engineering
- **Frameworks & Tools:** Django, Git, Unity, LaTeX, C#, .NET, Android Studio, dnSpy

HONORS AND AWARDS

Jason Lee Scholarship	August 2015 – May 2019
Willamette University College Honors	February 2016 – May 2019
Murdock Charitable Trust College Research Program for the Natural Sciences (\$5500)	June 2018 – August 2018